

ML Model based Optimisation of SNR Constrained Joint Beamforming and Reflection Designs for RIS-ISAC Systems

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Abstract—This paper investigates the integration of integrated sensing and communication (ISAC) and reconfigurable intelligent surfaces (RIS) to enable reliable communication and high-accuracy sensing. We consider an RIS-assisted ISAC system where a multi-antenna base station (BS) simultaneously performs multi-user communication and radar sensing. The joint design of transmit beamforming and RIS phase shifts is formulated to maximize the communication sum-rate under signal-to-noise ratio (SNR) constraint, subject to power and unit-modulus constraints. Due to the non-convex nature of the problem, conventional solutions rely on computationally expensive iterative algorithms. To address this, we propose a learning-based approach that replaces the optimization process with neural networks. Specifically, we generate training data using the original algorithms and train the Deep Neural Network (DNN) to learn the mapping from channel state information (CSI) to optimal beamforming and reflection parameters.

Index Terms—Integrated sensing and communication (ISAC), reconfigurable intelligent surface (RIS), multi-user multi-input, single-output (MU-MISO) communications, radar signal-to-noise ratio (SNR), Deep Neural Network (DNN), channel state information (CSI)

I. INTRODUCTION

Wireless communication and radar sensing have traditionally been developed as separate systems. Recently, integrated sensing and communication (ISAC) has emerged as a promising paradigm that enables both functionalities within a unified framework. This integration has evolved from coexistence to cooperation and further toward full co-design, enabled by similar hardware platforms and signal processing techniques, as well as the evolution toward high-frequency, wideband, and multi-antenna systems [1].

ISAC allows communication and sensing systems to share spectrum resources and transmit unified waveforms, thereby significantly improving spectral efficiency, energy efficiency, and hardware utilization [1]–[3]. It is expected to play a key role in next-generation wireless networks, supporting applications such as smart environments, autonomous systems, and vehicular networks that require both reliable communication and accurate sensing [4].

To enhance system performance, multi-input multi-output (MIMO) techniques have been widely investigated in ISAC systems. By exploiting spatial degrees of freedom, MIMO architectures improve communication performance through

beamforming and spatial multiplexing, while also enhancing sensing performance through waveform diversity [5]. However, despite these advantages, system performance may still degrade under unfavorable propagation conditions.

Recently, reconfigurable intelligent surface (RIS) technology has emerged as a promising solution to address these challenges. An RIS consists of a large number of passive reflecting elements whose electromagnetic responses can be intelligently controlled. By adjusting the phase shifts of these elements, RIS can reshape the wireless propagation environment, enhance signal strength, and provide additional degrees of freedom for system optimization [6]–[8].

Motivated by these advantages, RIS-assisted ISAC systems have attracted significant research interest. In such systems, the joint design of transmit beamforming at the base station and reflection coefficients at the RIS is crucial to achieving optimal performance. Existing work formulates this problem to maximize communication performance, such as sum-rate, while satisfying sensing constraints, including signal-to-noise ratio (SNR) for target detection [1].

However, the joint optimization of beamforming and RIS phase shifts is inherently non-convex due to the coupling between variables and the unit-modulus constraints of RIS elements. To solve this problem, prior works have proposed iterative optimization methods based on fractional programming (FP), majorization–minimization (MM), and alternating optimization techniques [1]. Although these approaches achieve near-optimal performance, they require multiple iterations for each channel realization, leading to high computational complexity and making them unsuitable for real-time implementation.

In the considered RIS-assisted ISAC framework, the optimal beamforming matrix and RIS phase shifts are obtained using iterative optimization algorithms [1]. These methods must be executed repeatedly for different channel realizations, resulting in significant computational overhead and latency. Therefore, there is a strong need for efficient approaches that can provide near-optimal solutions with reduced complexity.

A. Focus and Contributions

In this paper, we retain the original system model and problem formulation in [1], and focus on reducing computational complexity by replacing the optimization process with a learning-based approach. The main contributions are summarized as follows:

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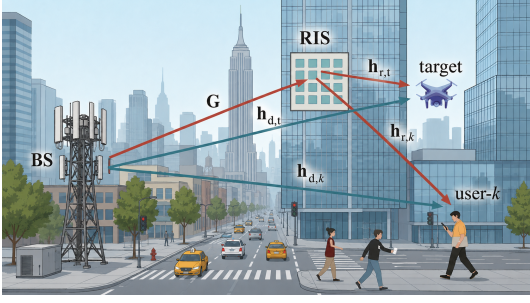


Fig. 1. An RIS-assisted ISAC system.

- We implement the optimization algorithms in [1] to generate a dataset of optimal beamforming matrices and RIS phase shifts for different channel realizations.
- We propose Deep Neural Network (DNN) architecture, to learn the mapping from channel state information (CSI) to optimal design parameters.
- We replace the iterative optimization procedure with direct inference using trained models, thereby significantly reducing computational complexity.
- We evaluate the proposed approach by reproducing performance matrix such as sum-rate under SNR constraint, and demonstrate that the learning-based models achieve performance comparable to the original optimization-based methods.

Notations: Boldface lower-case and upper-case letters indicate column vectors and matrices, respectively. $(\cdot)^*$, $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote the conjugate, transpose, conjugate transpose, and inverse operations, respectively. $\mathbf{I}_M$ indicates an $M \times M$ identity matrix. $\mathbb{C}$ and $\mathbb{R}$ denote the sets of complex numbers and real numbers, respectively. $|a|$, $\|\mathbf{a}\|$, and $\|\mathbf{A}\|_F$ are the magnitude of a scalar a , the norm of a vector $\mathbf{a}$, and the Frobenius norm of a matrix $\mathbf{A}$, respectively. $\mathbb{E}\{\cdot\}$ represents statistical expectation. $\text{Tr}\{\mathbf{A}\}$ takes the trace of the matrix $\mathbf{A}$ and $\text{vec}\{\mathbf{A}\}$ vectorizes the matrix $\mathbf{A}$. $\otimes$ denotes the Kronecker product. $\Re\{\cdot\}$ and $\Im\{\cdot\}$ denote the real and imaginary parts of a complex number, respectively. $\angle a$ is the angle of complex-valued a .

II. SYSTEM MODEL

We considered an RIS assisted monostatic ISAC system where a multi-antenna BS performs multi-user communication and radar sensing with the help of N elements RIS. BS is equipped with M transmit antennas and M receive antennas arranged with a uniform linear array (ULAs) with a spacing of half wavelength. Simultaneously serves K single-antenna users and senses one target with the addition of advanced self-interference mitigation techniques, it is assumed that BS operates in full-duplex mode with perfect self-interference mitigation. The focus is on two common and crucial radar sensing tasks: target detection, which aims to identify whether a target of interest is present, and parameter estimation, where the DoAs of the target signals with respect to the BS and the RIS are determined.

The dual functional transmitted signal for l -th slot is given by [5]

$$\mathbf{x}[l] = \mathbf{W}_c \mathbf{s}_c[l] + \mathbf{W}_r \mathbf{s}_r[l] = \mathbf{W} \mathbf{s}[l] \quad (1)$$

where $\mathbf{s}_c[l] \in \mathbb{C}^K$ denotes the communication symbol vector for K users with $\mathbb{E}\{\mathbf{s}_c[l] \mathbf{s}_c^H[l]\} = \mathbf{I}_K$. Similarly, $\mathbf{s}_r[l] \in \mathbb{C}^M$ represents the radar waveform with $\mathbb{E}\{\mathbf{s}_r[l] \mathbf{s}_r^H[l]\} = \mathbf{I}_M$ and $\mathbb{E}\{\mathbf{s}_c[l] \mathbf{s}_r^H[l]\} = \mathbf{0}_M$. The beamforming matrices for communication and radar are denoted by $\mathbf{W}_c \in \mathbb{C}^{M \times K}$ and $\mathbf{W}_r \in \mathbb{C}^{M \times M}$, respectively. For notational simplicity, we define the combined beamforming matrix $\mathbf{W} \triangleq [\mathbf{W}_c \ \mathbf{W}_r]$ and the stacked symbol vector $\mathbf{s}[l] \triangleq [\mathbf{s}_c^T[l] \ \mathbf{s}_r^T[l]]^T$.

The received signal at the k -th user can be expressed as

$$y_k[l] = (\mathbf{h}_{d,k}^T + \mathbf{h}_{r,k}^T \mathbf{\Phi} \mathbf{G}) \mathbf{x}[l] + n_k[l]. \quad (2)$$

where $\mathbf{h}_{d,k} \in \mathbb{C}^M$, $\mathbf{h}_{r,k} \in \mathbb{C}^N$, and $\mathbf{G} \in \mathbb{C}^{N \times M}$ denote the channels from the BS to the k -th user, from the RIS to the k -th user, and from the BS to the RIS, respectively. All channels are assumed to follow Rician fading. Specifically, the BS-RIS channel is modeled as

$$\mathbf{G} = \alpha_G \sqrt{\frac{\kappa}{\kappa + 1}} \mathbf{G}_{\text{LoS}} + \alpha_G \sqrt{\frac{1}{\kappa + 1}} \mathbf{G}_{\text{NLoS}}, \quad (3)$$

where α_G denotes the distance-dependent pathloss, κ is the Rician factor, $\mathbf{G}_{\text{LoS}}$ and $\mathbf{G}_{\text{NLoS}}$ represent the line-of-sight (LoS) and non-line-of-sight (NLoS) components, respectively. The NLoS component is modeled as Rayleigh fading with zero mean and unit variance.

The LoS component is given by

$$\mathbf{G}_{\text{LoS}} = \mathbf{a}_r^T(\theta_{\text{RB}}) \mathbf{a}_m^H(\theta_{\text{BR}}), \quad (4)$$

where $\mathbf{a}_N(\theta_{\text{RB}}) = [1, e^{-j\pi \sin \theta_{\text{RB}}}, \dots, e^{-j(N-1)\pi \sin \theta_{\text{RB}}}]^T$ and θ_{RB} , θ_{BR} denote the direct-of-arrival (DoA) and direct-of-departure (DoD), respectively.

We assume that accurate CSI estimates are assumed available at the BS, enabled by advanced estimation techniques for RIS-assisted systems. The RIS reflection matrix is defined as $\mathbf{\Phi} \triangleq \text{diag}\{\boldsymbol{\phi}\}$, where $\boldsymbol{\phi} = [\phi_1, \phi_2, \dots, \phi_N]^T$ with $|\phi_n| = 1$, $\forall n$. Finally, the additive noise at the k -th user is modeled as $n_k[l] \sim \mathcal{CN}(0, \sigma_k^2)$.

The transmitted signal propagates toward the target through both the direct and RIS-reflected links, and the reflected echo is subsequently received at the BS via these paths. Notably, the RIS operates in a passive full-duplex manner without introducing self-interference. Accordingly, the baseband echo signal received at the BS can be expressed as

$$y_r[l] = \alpha_t (\mathbf{h}_{d,t}^T + \mathbf{G}^T \mathbf{\Phi} \mathbf{h}_{r,t}) (\mathbf{h}_{d,t}^T + \mathbf{h}_{r,t}^T \mathbf{\Phi} \mathbf{G}) \mathbf{W} \mathbf{s}[l] + n_r[l]. \quad (5)$$

where $\alpha_t \sim \mathcal{CN}(0, \sigma_t^2)$ denotes the radar cross section (RCS). The channels $\mathbf{h}_{d,t} \in \mathbb{C}^M$ and $\mathbf{h}_{r,t} \in \mathbb{C}^N$ represent the BS-target and RIS-target links, respectively, while $n_r[l] \sim \mathcal{CN}(0, \sigma_r^2 \mathbf{I}_M)$ denotes the additive white Gaussian noise (AWGN).

Following standard radar assumptions, both BS–target and RIS–target channels are modeled as LoS links, given by

$$\mathbf{h}_{d,t} = \alpha_{dt} \mathbf{a}_M(\theta_1) \quad (6)$$

$$\mathbf{h}_{r,t} = \alpha_{rt} \mathbf{a}_N(\theta_2) \quad (7)$$

where θ_1 and θ_2 denote the angles of arrival (AoAs) of the target with respect to the BS and RIS, respectively. These angles can be estimated using conventional localization methods or maximum likelihood estimation based on prior observations.

Given θ_1 and θ_2 , the target detection and parameter estimation performance can be further enhanced via beamforming design at the BS. Moreover, due to the quasi-static nature of the BS–RIS channel, a two-timescale channel estimation framework can be employed to estimate $\mathbf{G}$.

Radar sensing is performed over L snapshots, which are stacked to form

$$\mathbf{Y}_r = \alpha_t \mathbf{H}_t(\phi) \mathbf{W} \mathbf{S} + \mathbf{N}_r, \quad (8)$$

where

$$\mathbf{H}_t(\phi) \triangleq (\mathbf{h}_{d,t} + \mathbf{G}^T \Phi \mathbf{h}_{r,t}) (\mathbf{h}_{d,t}^T + \mathbf{h}_{r,t}^T \Phi \mathbf{G}) \quad (9)$$

And the symbol and noise matrices are defined as

$$\mathbf{S} \triangleq [\mathbf{s}[1], \mathbf{s}[2], \dots, \mathbf{s}[L]] \quad (10)$$

$$\mathbf{N}_r \triangleq [n_r[1], n_r[2], \dots, n_r[L]] \quad (11)$$

Under the above assumptions, the equivalent channel $\mathbf{H}_t(\phi)$ is assumed to be known at the BS.

The objective in RIS-assisted ISAC systems is to jointly design the transmit beamforming matrix $\mathbf{W}$ and the RIS phase shift vector ϕ to maximize the communication sum-rate while satisfying sensing constraints such as signal-to-noise ratio (SNR) or Cramér–Rao bound (CRB), along with transmit power and unit-modulus constraints. This joint optimization problem is highly non-convex due to the coupling between $\mathbf{W}$ and ϕ and the unit-modulus constraint of RIS elements. Existing approaches solve this problem using iterative optimization algorithms, which incur high computational complexity and are not suitable for real-time implementation.

To address this issue, the problem is reformulated as a supervised learning task, where a neural network is trained to directly predict the RIS phase shifts from the channel state information (CSI).

The learned nonlinear mapping is represented as

$$\mathcal{F}_\Theta : \mathbf{X} \rightarrow \hat{\phi} \quad (12)$$

where $\mathbf{X}$ denotes the input CSI features, Θ represents the trainable network parameters, and $\hat{\phi}$ denotes the predicted RIS phase shift vector.

The input feature vector consists of the real and imaginary parts of the communication channels, while the network output is given by

$$\hat{\phi} = [e^{j\theta_1}, e^{j\theta_2}, \dots, e^{j\theta_N}]^T \quad (13)$$

where θ_n denotes the phase shift of the n -th RIS element.

III. PERFORMANCE METRICS

In RIS-assisted integrated sensing and communication (RIS-ISAC) systems, both communication and sensing functionalities are performed simultaneously using the same wireless resources. Therefore, suitable performance metrics are required to evaluate the communication quality and radar sensing capability of the considered system.

In this work, the achievable sum-rate is used as the communication performance metric, since it represents the maximum achievable data rate of all users in the network. For radar sensing, the target detection performance is characterized using the radar signal-to-noise ratio (SNR), which directly affects the probability of successful target detection.

The communication sum-rate and radar SNR are jointly influenced by the transmit beamforming matrix and RIS reflection coefficients. Hence, proper joint optimization of these parameters is necessary to achieve efficient communication performance while maintaining reliable sensing capability.

A. Communication Sum-Rate Analysis

From the received signal model, the effective SINR corresponding to the k -th communication user is expressed as

$$\text{SINR}_k = \frac{|\mathbf{h}_k^T(\phi) \mathbf{w}_k|^2}{\sum_{j=1, j \neq k}^{K+M} |\mathbf{h}_k^T(\phi) \mathbf{w}_j|^2 + \sigma_k^2}. \quad (14)$$

The equivalent BS-user channel including RIS reflection is represented as

$$\mathbf{h}_k(\phi) = \mathbf{h}_{d,k} + \mathbf{G}^T \Phi \mathbf{h}_{r,k}, \quad (15)$$

where $\mathbf{w}_j$ denotes the j -th beamforming vector of the combined transmit beamforming matrix

$$\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{K+M}]. \quad (16)$$

Using the above SINR formulation, the achievable sum-rate of all communication users can be obtained as

$$R = \sum_{k=1}^K \log_2(1 + \text{SINR}_k). \quad (17)$$

B. Radar SNR for Target Detection

For radar sensing, the target detection capability is evaluated through the received radar SNR. The received echo matrix after matched filtering can be represented as

$$\tilde{\mathbf{Y}}_r = \alpha_t \mathbf{H}_t(\phi) \mathbf{W} \mathbf{S} \mathbf{S}^H + \mathbf{N}_r \mathbf{S}^H. \quad (18)$$

By defining

$$\tilde{\mathbf{y}}_r = \text{vec}(\tilde{\mathbf{Y}}_r), \quad (19)$$

$$\mathbf{w} = \text{vec}(\mathbf{W}), \quad (20)$$

and

$$\tilde{\mathbf{n}}_r = \text{vec}(\mathbf{N}_r \mathbf{S}^H), \quad (21)$$

the vectorized received signal can be written as

$$\tilde{\mathbf{y}}_r = \alpha_t (\mathbf{S} \mathbf{S}^H \otimes \mathbf{H}_t(\phi)) \mathbf{w} + \tilde{\mathbf{n}}_r. \quad (22)$$

A receive beamforming vector $\mathbf{u} \in \mathbb{C}^{M \times (K+M)}$ is employed for radar processing, yielding

$$\mathbf{u}^H \tilde{\mathbf{y}}_r = \alpha_t \mathbf{u}^H (\mathbf{S} \mathbf{S}^H \otimes \mathbf{H}_t(\phi)) \mathbf{w} + \mathbf{u}^H \tilde{\mathbf{n}}_r. \quad (23)$$

The corresponding binary hypothesis testing problem is given by

$$y = \begin{cases} \mathcal{H}_0 : \mathbf{u}^H \tilde{\mathbf{n}}_r, \\ \mathcal{H}_1 : \alpha_t \mathbf{u}^H (\mathbf{S} \mathbf{S}^H \otimes \mathbf{H}_t(\phi)) \mathbf{w} + \mathbf{u}^H \tilde{\mathbf{n}}_r. \end{cases} \quad (24)$$

The Neyman-Pearson detector is formulated as

$$T = |y|^2 \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{\geq}} \delta, \quad (25)$$

where δ denotes the detection threshold.

The probabilities of false alarm and detection are expressed as

$$P_{\text{FA}} = \Pr(T > \delta | \mathcal{H}_0) = 1 - F_{\chi_2^2} \left(\frac{\delta}{\eta_0} \right), \quad (26)$$

$$P_{\text{D}} = \Pr(T > \delta | \mathcal{H}_1) = 1 - F_{\chi_2^2} \left(\frac{\delta}{\eta_1} \right), \quad (27)$$

where η_0 and η_1 are the variances under hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively.

For a fixed false alarm probability, the achieved detection probability becomes

$$P_{\text{D}} = 1 - F_{\chi_2^2} \left(\frac{\eta_0}{\eta_1} F_{\chi_2^2}^{-1}(1 - P_{\text{FA}}) \right). \quad (28)$$

Therefore, the detection performance is directly related to the radar SNR, which can be expressed as

$$\text{SNR}_t = \frac{\sigma_t^2 \mathbb{E} \left\{ |\mathbf{u}^H (\mathbf{S} \mathbf{S}^H \otimes \mathbf{H}_t(\phi)) \mathbf{w}|^2 \right\}}{L \sigma_r^2 \mathbf{u}^H \mathbf{u}}. \quad (29)$$

To simplify the optimization framework, a lower bound on the radar SNR is utilized through Jensen's inequality:

$$\text{SNR}_t \geq \frac{L \sigma_t^2 |\mathbf{u}^H (\mathbf{I}_{K+M} \otimes \mathbf{H}_t(\phi)) \mathbf{w}|^2}{\sigma_r^2 \mathbf{u}^H \mathbf{u}}. \quad (30)$$

This lower bound corresponds to the worst-case radar sensing performance

C. ML-Based Performance Evaluation

The generated optimization dataset is further utilized for training a machine learning model to predict RIS reflection coefficients and beamforming-related parameters. The predicted parameters are subsequently used for effective channel construction and achievable sum-rate evaluation under different transmit power levels.

The communication performance of the trained model is analyzed using the achievable sum-rate versus transmit power characteristics. The obtained results are further compared with the optimization-based and benchmark schemes presented in the reference work.

The ML-based framework avoids repeated iterative optimization during online implementation and enables faster system-level performance evaluation with reduced computational complexity.

IV. PROPOSED SOLUTION

This section presents the proposed machine learning assisted RIS-ISAC optimization framework for intelligent beamforming and adaptive RIS phase configuration. The proposed methodology aims to learn the nonlinear relationship between the wireless channel environment and the corresponding optimized RIS configuration using a deep neural network (DNN). Unlike conventional iterative optimization approaches, the proposed framework performs offline learning and enables low-latency real-time inference during deployment.

The overall framework consists of dataset generation, feature preprocessing, deep neural network based optimization, RIS phase reconstruction, beamforming design, and achievable sum-rate evaluation. The proposed approach significantly reduces the computational burden associated with iterative RIS optimization while maintaining efficient communication performance.

A. RIS Reflection and Effective Channel Modeling

Consider an RIS-assisted multi-user communication system where the RIS is composed of multiple passive reflecting elements capable of independently adjusting the phase of the incident electromagnetic wave. The RIS reflection matrix is represented as

$$\Phi = \text{diag}(e^{j\theta_1}, e^{j\theta_2}, \dots, e^{j\theta_N}) \quad (31)$$

where N denotes the number of RIS elements and θ_n represents the phase shift introduced by the n^{th} RIS element.

Since the RIS operates passively, the reflection coefficients satisfy the unit modulus constraint given by

$$|\phi_n| = 1, \quad \forall n = 1, 2, \dots, N \quad (32)$$

The RIS dynamically modifies the wireless propagation environment by intelligently adjusting the phase shifts of the reflected signals. Proper RIS configuration enables constructive signal combining at the receiver and improves the achievable communication performance. The overall communication link

consists of both direct and RIS-assisted reflected propagation paths. Let $\mathbf{H}_d$ denote the direct BS-user channel, $\mathbf{G}$ denote the BS-RIS channel, and $\mathbf{H}_r$ denote the RIS-user channel.

The effective RIS-assisted communication channel can be expressed as

$$\mathbf{H}_{eff} = \mathbf{H}_d + \mathbf{H}_r \Phi \mathbf{G} \quad (33)$$

The RIS phase matrix Φ intelligently modifies the reflected signal components to enhance the equivalent channel gain and suppress unfavorable propagation effects.

B. Dataset Generation

The proposed framework employs supervised learning to approximate the complex optimization mapping between wireless channel conditions and optimized RIS configurations. To enable effective training of the neural network, a dataset is constructed using multiple wireless channel realizations generated under different communication scenarios. Each dataset sample consists of input channel features and their corresponding optimized RIS-related target variables obtained through optimization procedures. The generated dataset captures the variations in wireless propagation environments and enables the neural network to learn generalized optimization behavior.

The input feature space includes channel state information associated with the direct and RIS-assisted communication links. In addition, power-related features are incorporated to improve the adaptability of the proposed framework under varying transmit power conditions. The target labels correspond to optimized RIS reflection coefficients and beamforming-related optimization variables. By learning this nonlinear mapping, the proposed neural network can directly predict optimized RIS configurations during inference without requiring iterative optimization.

1) *Data Normalization*: Since the channel features may possess significantly different numerical ranges, data normalization is performed prior to training to improve convergence stability and learning efficiency.

The normalized feature representation is expressed as

$$X_{norm} = \frac{X - \mu_X}{\sigma_X} \quad (34)$$

where μ_X and σ_X represent the mean and standard deviation of the input features, respectively.

Normalization prevents feature dominance during optimization and improves the numerical stability of the learning process.

2) *Feature Augmentation Strategy*: To improve the adaptability of the proposed framework across different operating conditions, the transmit power information is incorporated into the input feature vector. This enables the neural network to learn power-aware RIS optimization strategies.

The augmented feature representation is given by

$$\tilde{X} = [X, P_t] \quad (35)$$

TABLE I
FUNCTIONAL DESCRIPTION OF THE PROPOSED NEURAL NETWORK ARCHITECTURE

Layer Type	Function
Input Layer	Receives normalized channel features
Dense Layers	Extract nonlinear feature representations
Batch Normalization	Stabilizes network training
Dropout Layer	Prevents overfitting
Output Layer	Predicts optimization variables

TABLE II
ARCHITECTURE OF THE PROPOSED DEEP NEURAL NETWORK

Layer	Activation	Output Dimension	Trainable Parameters	Complexity
Input	–	n_{in}	0	$\mathcal{O}(1)$
Dense-1	LeakyReLU	2048	$(n_{in} + 1) \times 2048$	$\mathcal{O}(n_{in} \times 2048)$
BatchNorm-1	–	2048	4096	$\mathcal{O}(2048)$
Dropout-1	–	2048	0	$\mathcal{O}(2048)$
Dense-2	LeakyReLU	1024	$(2048 + 1) \times 1024$	$\mathcal{O}(2048 \times 1024)$
BatchNorm-2	–	1024	2048	$\mathcal{O}(1024)$
Dropout-2	–	1024	0	$\mathcal{O}(1024)$
Dense-3	LeakyReLU	512	$(1024 + 1) \times 512$	$\mathcal{O}(1024 \times 512)$
Dense-4	LeakyReLU	256	$(512 + 1) \times 256$	$\mathcal{O}(512 \times 256)$
Output	Linear	n_{out}	$(256 + 1) \times n_{out}$	$\mathcal{O}(256 \times n_{out})$

where X denotes the original channel feature vector and P_t denotes the transmit power feature.

The inclusion of transmit power information enables the neural network to generalize across multiple power conditions and improves the robustness of the learned optimization framework.

C. Deep Neural Network Based Optimization

The proposed optimization framework employs a fully connected deep neural network to learn the nonlinear mapping between the wireless channel features and optimized RIS configurations. The neural network performs hierarchical nonlinear feature extraction using multiple dense layers. The extracted features progressively capture complex channel relationships and optimization characteristics associated with RIS-assisted communication. LeakyReLU activation functions are utilized to improve nonlinear learning capability and avoid the dying ReLU problem. Batch normalization layers are incorporated to stabilize training and accelerate convergence, while dropout regularization is employed to reduce overfitting and improve generalization performance.

Conventional RIS optimization approaches generally rely on iterative optimization procedures involving repeated matrix inversion and alternating optimization. Such approaches often introduce substantial computational complexity and optimization latency, particularly for large-scale RIS systems. In contrast, the proposed framework shifts the optimization burden to an offline training stage. During inference, the trained neural network directly predicts optimized RIS configurations using forward propagation operations.

The computational complexity of a fully connected layer is primarily governed by matrix multiplication operations and can be approximated as

$$\mathcal{O}(n_{in} \times n_{out}) \quad (36)$$

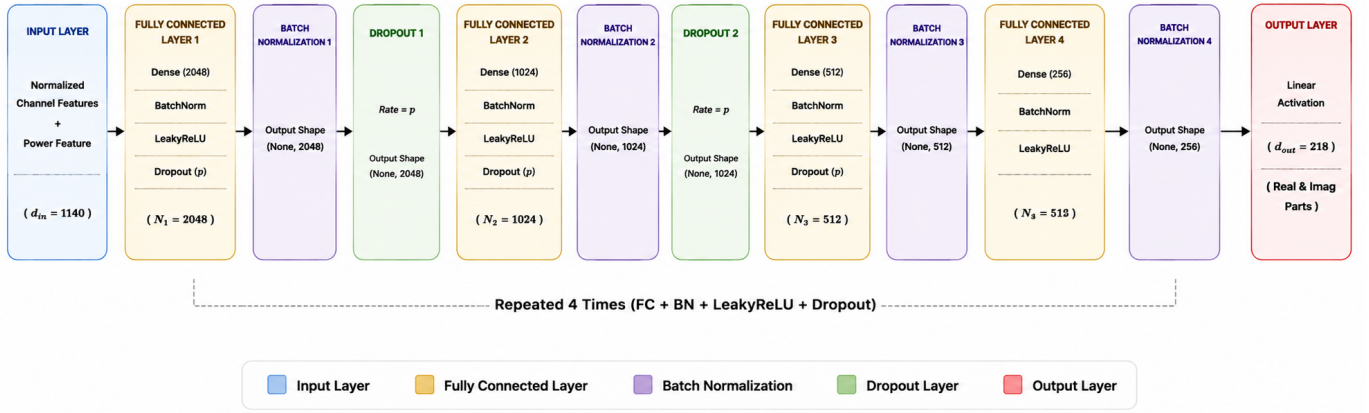


Fig. 2. Architecture of the proposed deep learning-based RIS optimization framework.

where n_{in} and n_{out} denote the number of input and output neurons, respectively.

1) *Training Objective and Loss Function:* The neural network is trained as a supervised regression model using the mean squared error (MSE) loss function. The objective of the training process is to minimize the prediction error between the predicted and optimized RIS variables. The MSE loss function is defined as

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (37)$$

where y_i and $\hat{y}_i$ denote the target and predicted optimization variables, respectively. The optimization process updates the trainable network parameters iteratively to minimize the overall prediction error.

2) *RIS Reconstruction:* The neural network predicts the real and imaginary components associated with the RIS reflection coefficients. The complex RIS coefficients are reconstructed as

$$\phi_n = \phi_{real,n} + j\phi_{imag,n} \quad (38)$$

To satisfy the passive RIS constraint, unit modulus normalization is enforced as

$$\frac{\phi_n}{|\phi_n|} \rightarrow \phi_n \quad (39)$$

This normalization projects the predicted RIS coefficients onto the feasible RIS reflection space.

3) *Zero forcing Beamforming:* Zero-forcing (ZF) beamforming is a linear precoding technique commonly employed in multi-user MIMO communication systems to suppress inter-user interference. In conventional multi-user transmission, signals intended for different users interfere with each other due to simultaneous transmission over the same wireless channel. The primary objective of ZF beamforming is to eliminate this interference by designing mutually orthogonal transmit beamforming vectors.

In the ZF beamforming framework, the transmit beamforming matrix is constructed using the effective channel matrix of the system. The ZF beamforming matrix is expressed as

$$W = H^H (H H^H)^{-1} \quad (38)$$

where H represents the effective RIS-assisted channel matrix. The matrix inverse operation approximately diagonalizes the multi-user channel, thereby reducing interference among users.

In the proposed work, the RIS reflection coefficients predicted by the neural network are first utilized to reconstruct the RIS-assisted effective channel. The reconstructed channel contains both the direct BS-user channel component and the RIS-reflected channel component. After obtaining the effective channel matrix, ZF beamforming is applied to generate the transmit beamforming matrix corresponding to different users.

The generated ZF beamforming vectors orthogonalize the transmitted user signals, which significantly reduces multi-user interference and improves the achievable sum-rate performance. The combination of RIS-assisted channel reconstruction and ZF beamforming enables efficient communication performance under different transmit power levels while maintaining low computational complexity.

4) *Achievable Rate Analysis:* The communication performance of the proposed framework is evaluated using the achievable sum-rate metric. The received signal quality for the k^{th} user is characterized by the signal-to-interference-plus-noise ratio (SINR) expressed as

$$\text{SINR}_k = \frac{|h_k^H w_k|^2}{\sum_i |h_k^H w_i|^2 + \sigma^2} \quad (40)$$

where h_k denotes the effective channel associated with the k^{th} user, w_k denotes the beamforming vector, and σ^2 represents the noise variance.

The achievable rate of the k^{th} user is computed as

$$R_k = \log_2(1 + \text{SINR}_k) \quad (41)$$

Accordingly, the overall system sum-rate is given by

$$R_{\text{sum}} = \sum_{k=1}^K R_k \quad (42)$$

V. NUMERICAL RESULTS

The system parameters are selected as $M = 6$, $K = 4$, $\sigma_k^2 = \sigma_t^2 = -90$ dBm $\forall k$, and $L = 1024$. The distances corresponding to the BS-RIS, RIS-target, and RIS-user links are considered as $d_{\text{BR}} = 50$ m, $d_{\text{RT}} = 3$ m, and $d_{\text{RU}} = 8$ m, respectively. In addition, the angular parameters are chosen as $\theta_{\text{BR}} = \theta_2 = \frac{\pi}{4}$, $\theta_{\text{RB}} = \frac{\pi}{4}$, while the target angle is determined by $\theta_1 = \tan^{-1}\left(\frac{d_{\text{BR}} \sin \theta_{\text{BR}} + d_{\text{RT}} \sin \theta_2}{d_{\text{BR}} \cos \theta_{\text{BR}} + d_{\text{RT}} \sin \theta_2}\right)$. Accordingly, the separation distances between the BS-target and BS-user links can be obtained geometrically.

The path-loss exponents for the BS-RIS, RIS-target, RIS-user, BS-target, and BS-user channels are set to 2.2, 2.2, 2.3, 2.4, and 3.5, respectively. Since the users are assumed to be significantly farther from the target location, the reflected components from the target towards the users are neglected due to severe attenuation effects. Furthermore, the Rician fading factors associated with the BS-RIS/user and RIS-user links are fixed at $\kappa = 3$ dB. The RIS phase-shift matrix is modeled using a unit-modulus constraint with random phase realizations.

The simulations are performed over a transmit power range of 27 to 32 dBm to evaluate the achievable sum-rate performance under different operating conditions. The proposed deep neural network is trained using the Adam optimizer with a learning rate of 10^{-3} to ensure stable and efficient convergence during optimization. A batch size of 64 and 50 training epochs are employed throughout the training process. To improve the generalization capability of the proposed framework and reduce overfitting, a dropout rate of 0.2 is incorporated within the network architecture. The generated dataset consists of 3000 samples obtained from multiple wireless channel realizations. Among these samples, 80% of the dataset is utilized for network training, while the remaining 20% is reserved for testing and performance evaluation.

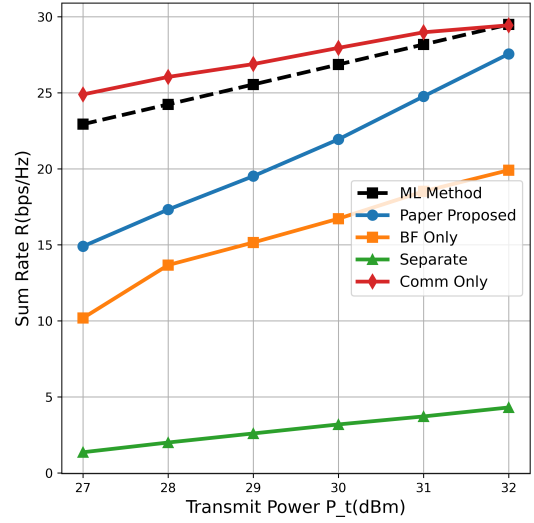


Fig. 3. Achievable sum-rate performance VS transmit power for different optimization schemes.

A. Performance Comparison of Different Methods

Fig. 3 compares the achievable sum-rate performance of the proposed ML-based optimization framework with the paper proposed scheme, beamforming-only method, communication-only strategy, and separate optimization approach under different transmit power levels.

It can be observed that the proposed ML method achieves significantly higher sum-rate performance than the beamforming-only and separate optimization schemes across the entire transmit power range. The performance improvement is achieved through intelligent learning of RIS phase configurations and adaptive beamforming optimization using the trained deep neural network.

Compared with the paper proposed scheme, the proposed ML framework demonstrates improved sum-rate performance at all transmit power levels. This improvement indicates that the learning-based optimization strategy is capable of efficiently capturing the nonlinear relationship between the wireless channel conditions and optimized RIS configurations.

Although the communication-only scheme achieves the highest communication rate, it allocates all available resources solely for communication transmission without considering integrated sensing requirements. In contrast, the proposed ML framework provides a balanced optimization strategy while still maintaining competitive communication performance. Furthermore, the achievable sum-rate of the proposed ML method increases consistently with increasing transmit power, demonstrating the robustness and scalability of the proposed framework under different operating conditions.

VI. CONCLUSIONS

In conclusion, this work proposed a deep learning-based framework for RIS-assisted wireless communication systems to reduce the complexity of conventional optimization methods. A DNN was used to predict RIS phase shifts directly

from channel state information, followed by Zero-Forcing beamforming for communication enhancement. Simulation results demonstrated improved achievable sum-rate performance under different transmit power levels while significantly reducing computational complexity. The results show that machine learning can provide an efficient and scalable solution for RIS-assisted communication systems.

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